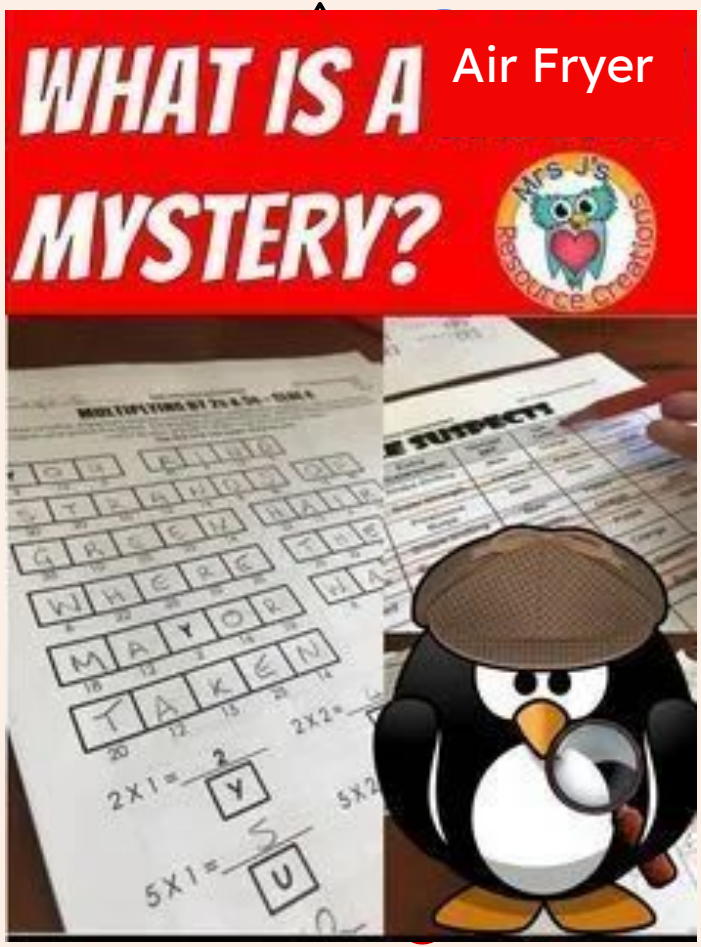
Final Project ML

Overview

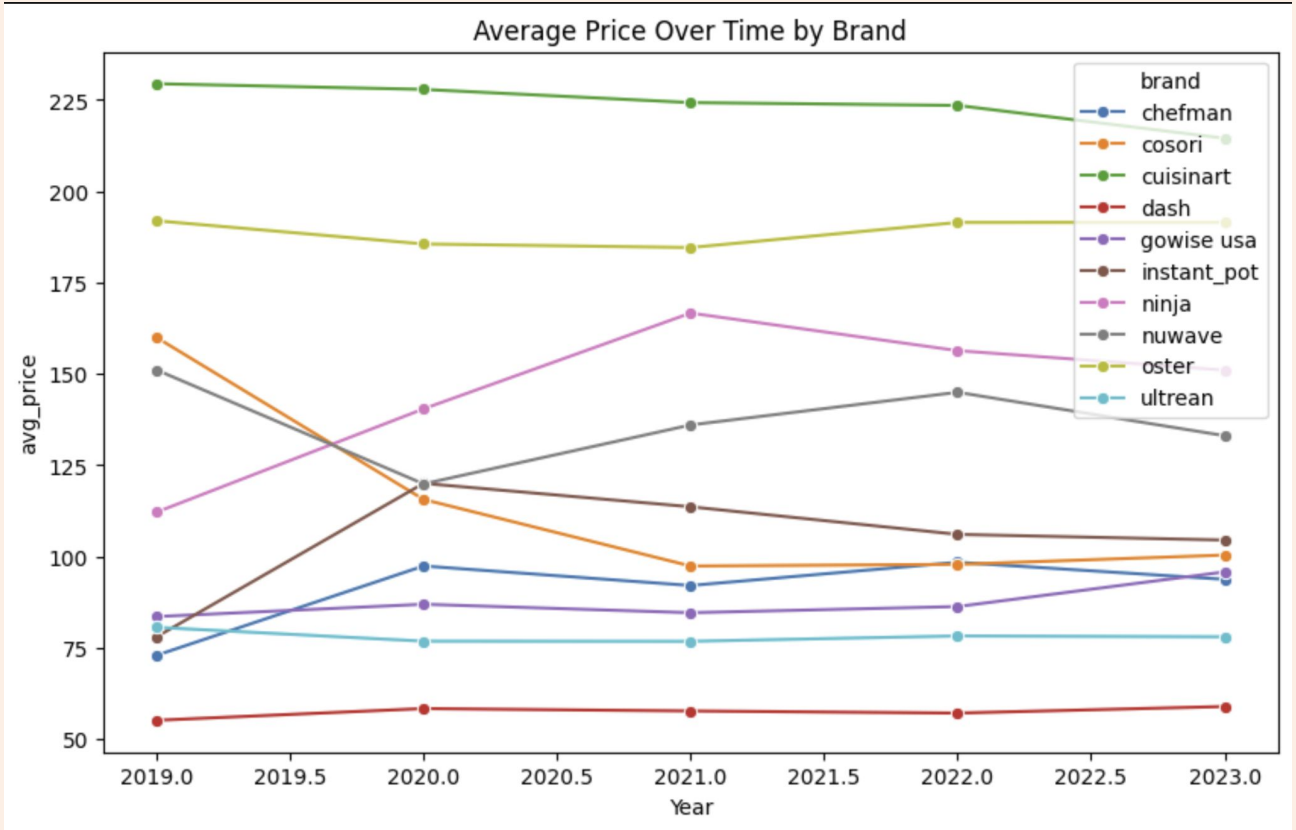
Our group has just begun to unravel the deep complexities of the air fryer market. So far, we have seen brands fighting for dominance, but only our reasearch will shed light on this never-ending mystery



The Air Fryer Market Became Increasingly Competitive (2019–2023)

- Market consists of 10 major brands on Amazon (2019–2023)
- Demand measured using purchase share by brand
- Market shifted from early leaders to stronger dominant brands
- Analysis focuses on price, ratings, features, and profitability

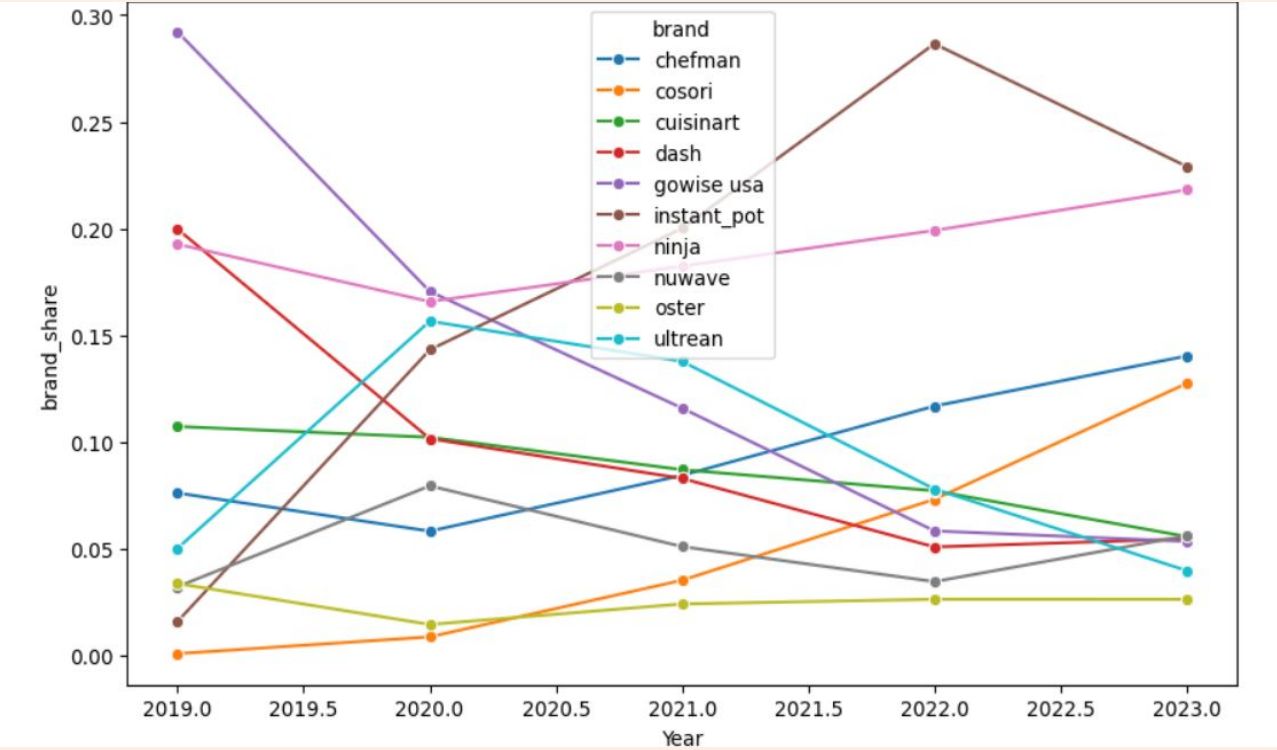
Average Brand Price Over Time



Clear Pricing Tiers Exist, With Limited Volatility Over Time

- Cuisinart and Oster consistently price at the premium tier (~$190–$230)
- Dash and Ultrean remain low-cost options (~$55–$80)
- Most brands show stable pricing with only gradual adjustments over time

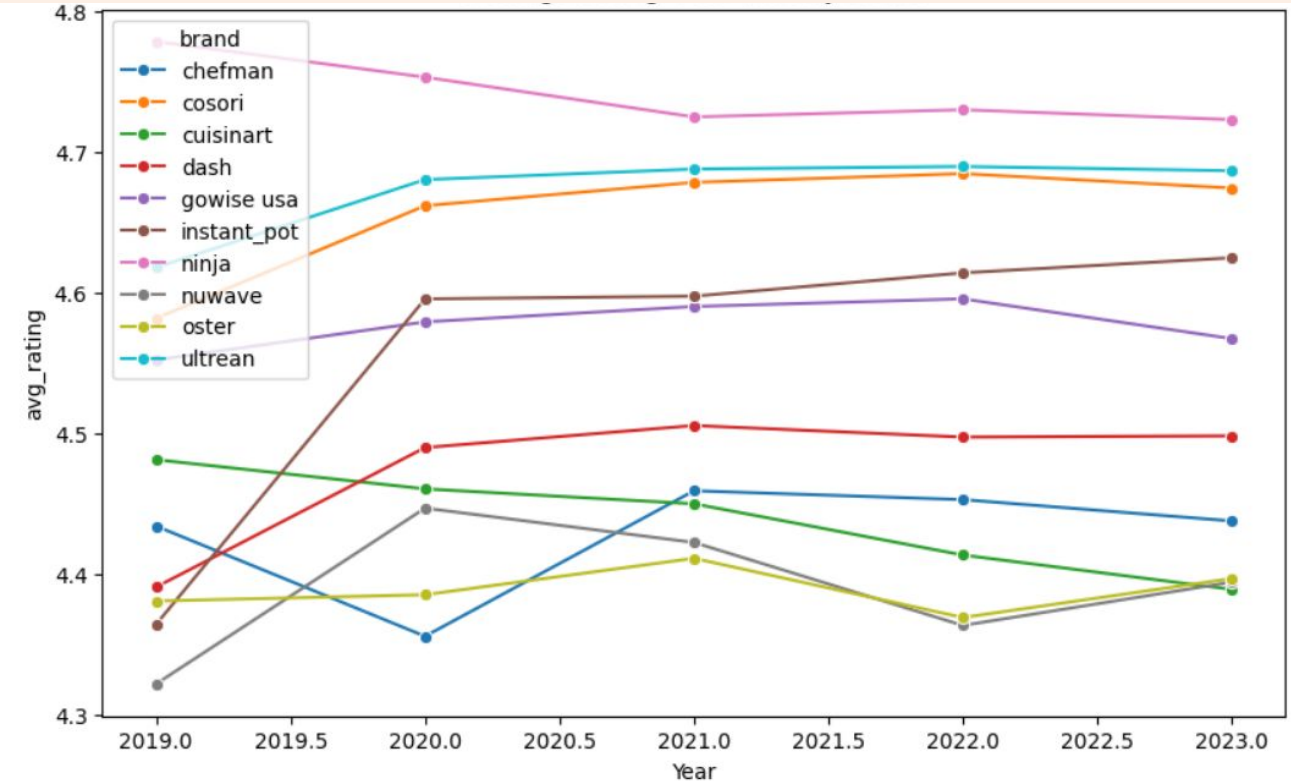
Brand Share Over Time



Market Share Shifted Toward New Leading Brands Over Time

- GoWISE USA started as the market leader but saw a significant decline over time
- Ninja and Instant Pot emerged as the dominant brands by 2022–2023
- Cosori and Chefman show steady growth, while other brands remain relatively stable

Brand Rating Over Time



Customer Ratings Remain High, With Limited Differentiation Across Brands

- All brands maintain consistently high ratings (~4.4–4.7)
- Ninja and Ultrean rank among the highest-rated brands over time
- Ratings show minimal variation, suggesting quality is similar across competitors

Modeling Consumer Choice in the Air Fryer Market

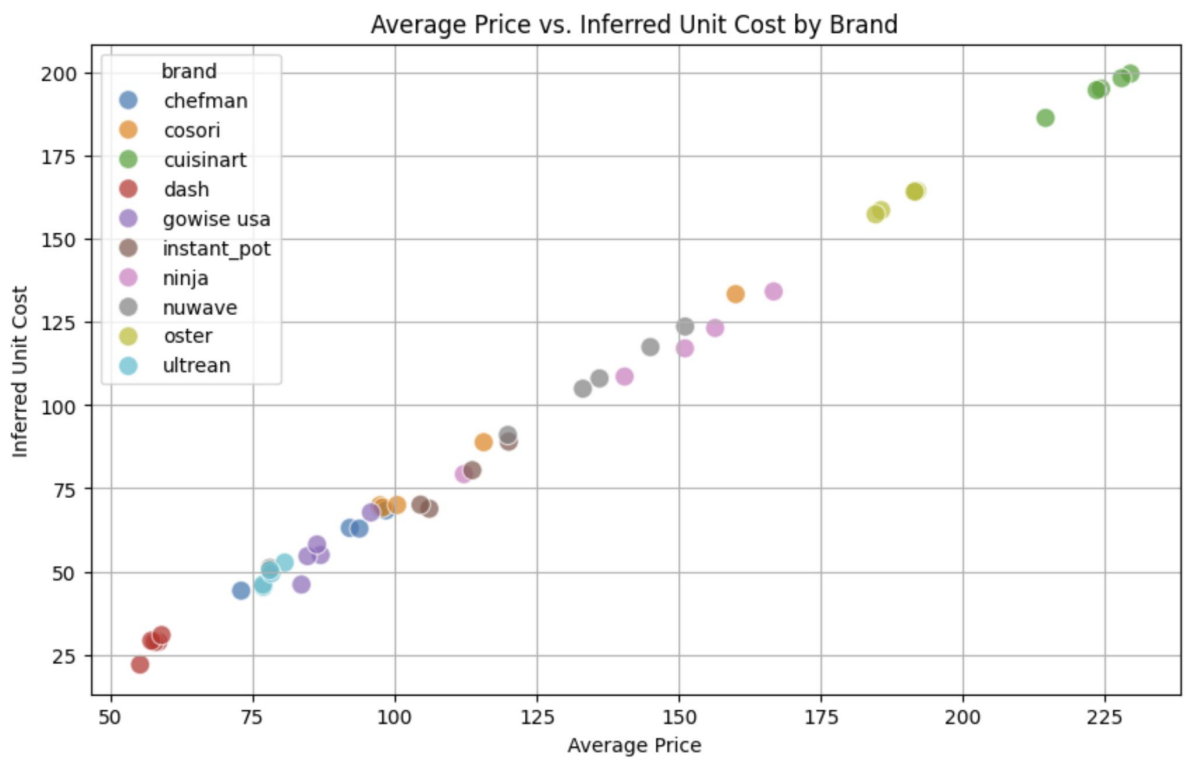
- We estimate demand using a logit-style regression model of market share
- The dependent variable is brand share, explained by:
  - Price
  - Customer ratings
  - Product features (e.g., design types)
- We include brand fixed effects to capture inherent brand preference
- We include year fixed effects to capture changes in demand over time
- This framework allows us to quantify how each factor impacts demand

$$\log(s_{bt}) = \alpha_0 + \alpha_t + \gamma_b + \beta_{price} p_{bt} + \beta_{rating} r_{bt} + \sum_{\ell=1}^{L} \beta_\ell x_{bt\ell} + \epsilon_{bt}.$$

Price Has a Significant Negative Impact on Demand

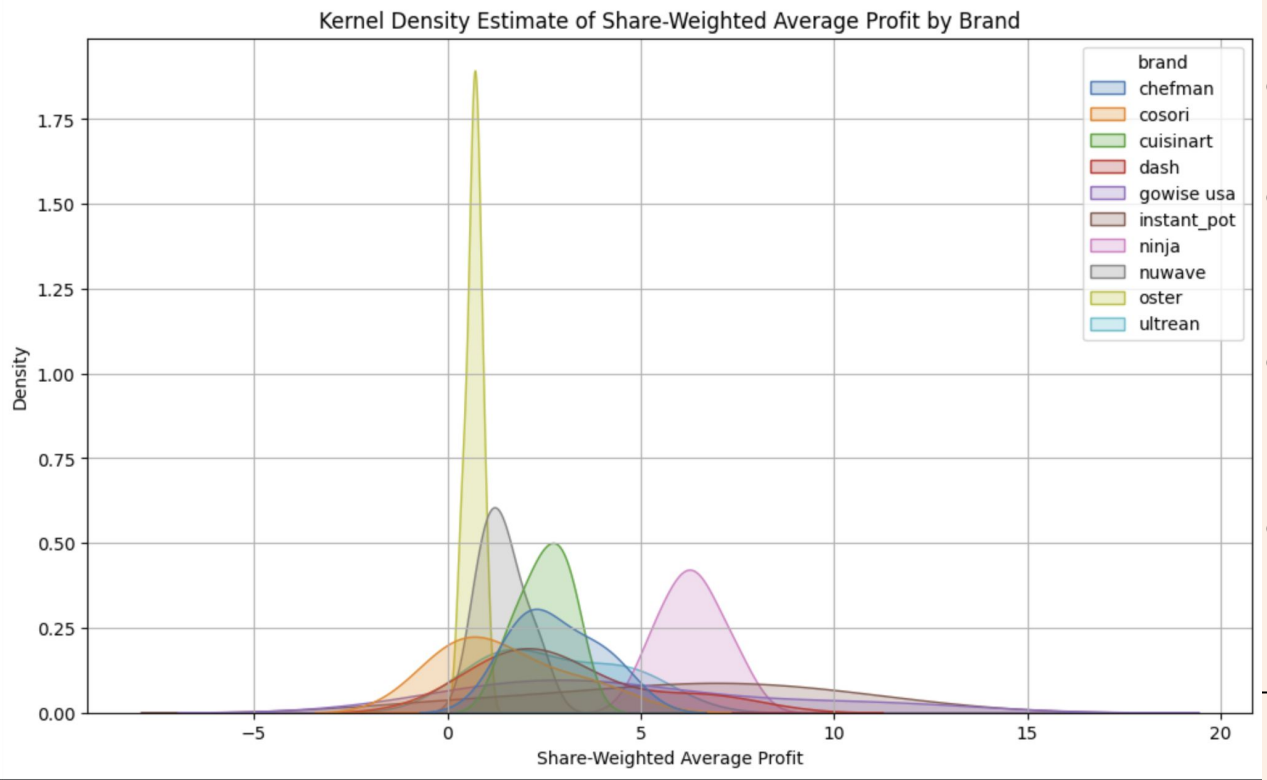
- The estimated price coefficient is −0.0377, indicating a negative relationship between price and demand
- As prices increase, a brand’s market share declines, holding other factors constant
- This suggests consumers are sensitive to price differences when choosing between competing brands
- Even with relatively stable pricing across brands, small differences still influence purchasing decisions
- This result is consistent with the law of demand, confirming expected economic behavior

Cost Structures and Markups Vary Across Brands



- Premium brands (e.g., Cuisinart, Oster) have the highest unit costs
- Lower-cost brands (e.g., Dash, Ultrean) operate with significantly lower costs
- Markups are relatively consistent across brands (~27–33)
- Higher prices are generally aligned with higher underlying costs

Profitability Depends on Both Margins and Market Share



- Share-weighted profit varies significantly across brands
- Brands like Ninja, GoWise and Instant Pot show strong profitability
- Some premium brands have high costs but not proportionally higher profits
- No negative unit costs → model results are economically consistent

Conclusion

- Ninja leads the market in consistent profitability and strong demand
- Instant Pot shows high upside with strong but more variable performance
- Premium brands (e.g., Cuisinart, Oster) have high costs but not proportionally higher profits
- Price matters, but product features and brand strength drive differentiation
- Overall, success depends on balancing cost, demand, and positioning

Members

Alexander Scott : Worked on Data Analysis in the workbook. Did the brand Share/Price/Rating over time slides

Safaa Mahbub: Worked on the demand estimation (part 2); Worked on the explanation of the demand model slide

Sami Kang: Successfully completed Part 3, verifying that all data aligns with the overall project targets.

Logan Lee: Worked on converting the data into a slideshow